

# SARAH LIU

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## TECHNICAL SKILLS

**Certifications:** Google AI, Google UX Design, Cisco Networking Basics  
**Languages:** Python, Java, JavaScript, TypeScript, Swift, SQL, C++  
**Frameworks & Libraries:** FastAPI, Node.js, REST APIs, JWT, OAuth, React, Next.js, SwiftUI, Tailwind CSS, Zustand  
**Databases:** PostgreSQL, Supabase, MongoDB, Neon, Row Level Security  
**DevOps & Tools:** Git, Docker, Azure DevOps, CI/CD, Vercel, Linux, Bash, Stripe  
**AI Tooling:** Claude Code, Cursor, GitHub Copilot, Codex, Gemini, v0 by Vercel, Groq, GPT-4

## PROJECTS

- Oat – Productivity & Planning SaaS** | [OatPlanner.com](#) Mar 2026 – Present  
*Next.js, TypeScript, Supabase, PostgreSQL, Stripe, Groq* Remote
- Sole engineer for a production SaaS with **100+ users**; built a **Next.js/TypeScript** planning app with calendar APIs, recurring events, voice scheduling, habit tracking, and **Zustand**-managed client state.
  - Designed a custom **JWT** authentication and authorization flow on **Supabase/PostgreSQL**; enforced Row Level Security on user data and added middleware security headers and rate limiting for API routes.
  - Integrated Google Calendar **OAuth**, **Stripe** subscriptions, and a **Groq**-hosted LLM behind server-side handlers, keeping billing, scheduling, and AI features cleanly separated.
  - Owned the full **software development lifecycle** from schema design and API implementation through deployment on **Vercel**, with ongoing fixes and improvements based on real user feedback.
- ASML-Affiliated Tabletop Cleanroom – Telemetry Dashboard** | [Project ISO](#) Sept 2025 – May 2026  
*Arduino C++, JavaScript, Web Serial API, Chart.js, Tailwind CSS* Fairfield, CT
- Technical lead for a 3-engineer senior design team; designed a dual-microcontroller **Arduino C++** architecture that separated telemetry collection from motion control, eliminating sensor polling bottlenecks.
  - Wrote firmware aggregating live readings from **4** environmental sensors at **10 Hz** and streamed data over the **Web Serial API** to a **Chart.js** dashboard at under **200ms** end-to-end latency.
  - Delivered real-time ISO 14644-1 compliance telemetry that certified the chamber as ISO Class 6, earning the **Dean’s Award for Best Senior Design** and **2nd Place** at the SME Chapter 7 Competition.
- Sportly – Drop-In Sports Event Platform** | In Development June 2026 – Present  
*Next.js 14, PostgreSQL (Neon), Drizzle ORM, Auth.js v5, Stripe Connect* Remote
- Co-building an event registration platform that works for any sport, using **Next.js 14** with **Stripe Connect** for direct organizer payouts and **Auth.js v5** magic-link authentication with guest checkout.
  - Designed a transactional **PostgreSQL** capacity reservation system preventing overselling by temporarily holding spots during checkout and auto-releasing expired holds, backed by idempotent **Stripe** webhook processing for reliable payment reconciliation.

## EXPERIENCE

- Sai – Software Engineering Intern** May 2025 – Aug 2025  
*Next.js, GPT-4o, OAuth2.0, Stripe, React* Remote
- Joined an existing multimodal chatbot codebase; built, enhanced, and tuned **GPT-4o API** prompts for student queries, diagram generation, and flashcard/quiz creation, and added **Google Auth OAuth2.0** sign-in and **Stripe** integration for session-based chat history and user verification.
  - Improved the multilingual **React/CSS** interface with responsive, cross-device fixes and pre-deploy testing on **Vercel**.
  - Reviewed and fixed bugs in existing code, working directly with the **CTO** on sprint planning and ticket-based task prioritization.
- Fairfield University – Teaching Assistant (Cloud Computing)** Jan 2026 – May 2026  
*Azure DevOps, Linux, Bash* Fairfield, CT
- Supported **30+** students on **Azure DevOps CI/CD** pipelines, **Linux**, Bash, and containerization labs; identified recurring blockers during grading and helped reduce repeat errors across lab cohorts.

## LEADERSHIP

- Team Captain | Fairfield Women’s Tennis – NCAA Division I** Sept 2025 – May 2026  
*Fairfield University Athletics* Fairfield, CT
- Primary liaison across athletes, coaching staff, and athletic department for a 10-person D1 roster, managing competing priorities across 25+ hrs/week of practice and travel.
  - Delivered the match-clinching win in a 3-3 tied team match to win the 2024 **MAAC Conference Championship** and qualify for the **NCAA Tournament**.
- sliu.log | Technical Content Brand & Community** Sept 2025 – Present  
*Adobe Creative Suite, Audience Analytics, Short-Form Video* Remote
- Grew a technical content brand to **10,000+ followers** (3.5% avg. engagement, 2–3× industry benchmark); launched a Chicago-based Discord growing **0 → 400 members in 1 week**.

## EDUCATION

- Fairfield University** May 2026  
*Bachelor of Science in Computer Science, Minor in Mathematics | Honors Program* Fairfield, CT
- GPA: 3.80/4.00 | Dean’s List 2022–2026 | Upsilon Pi Epsilon Computer Science Honor Society
  - Relevant Coursework: Data Structures, Algorithms, Database Systems, Operating Systems, Networks, Object-Oriented Programming, Computer Architecture, Discrete Mathematics, Data Science